FRONT WHEEL SPEED SENSOR

DESCRIPTION

A speed sensor is used at each wheel. The front sensors are mounted to the steering knuckles. The rear sensors are mounted to the rear brake backing plate.

OPERATION

The sensors convert wheel speed into a small AC electrical signal. This signal is transmitted to the CAB. The CAB convert the AC signal into a digital signal for each wheel. This voltage is generated by magnetic induction when a tone wheel passes by the stationary magnetic of the wheel speed sensor.

A gear type tone ring serves as the trigger mechanism for each sensor. The tone rings are mounted at the outboard ends of the front and rear axle shafts.

Different sensors are used at the front and rear wheels (Fig. 1). The front/rear sensors have the same electrical values but are not interchangeable. The sensors have a resistance between 900 and 1300 ohms.

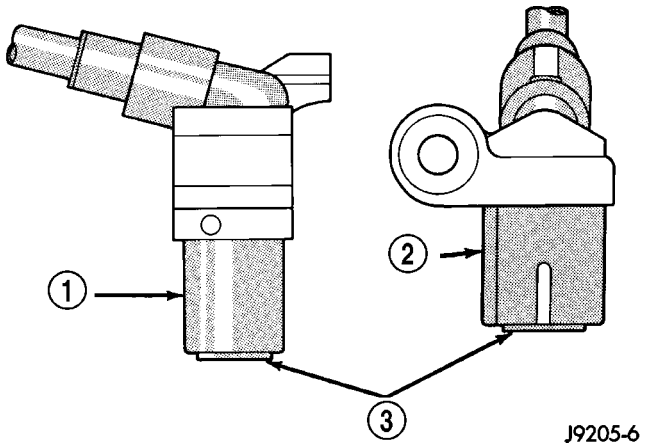

Fig. 1 Typical Wheel Speed Sensors

1 - FRONT SENSOR
2 - REAR SENSOR
3 - PICKUP FACE

FRONT SENSOR AIR GAP

Front sensor air gap is fixed and not adjustable. Only rear sensor air gap is adjustable.

Although front air gap is not adjustable, it can be checked if diagnosis indicates this is necessary. Front air gap should be 0.40 to 1.3 mm (0.0157 to 0.051 in.). If gap is incorrect, the sensor is either loose, or damaged.

REAR SENSOR AIR GAP

A rear sensor air gap adjustment is only needed when reinstalling an original sensor. Replacement sensors have an air gap spacer attached to the sensor pickup face. The spacer establishes correct air gap when pressed against the tone ring during installation. As the tone ring rotates, it peels the spacer off the sensor to create the required air gap. Rear sensor air gap is 0.28-1.5 mm (0.011-0.059 in.).

Sensor air gap measurement, or adjustment procedures are provided in this section. Refer to the front, or rear sensor removal and installation procedures as required.

REMOVAL

(1) Raise vehicle and turn wheel outward to access the sensor.

(2) Disconnect sensor wire connector at harness plug.

(3) Remove sensor wire from mounting retainers.

(4) Clean sensor and surrounding area with shop towel before removal.

(5) Remove bolt attaching sensor to steering knuckle and remove sensor (Fig. 2).

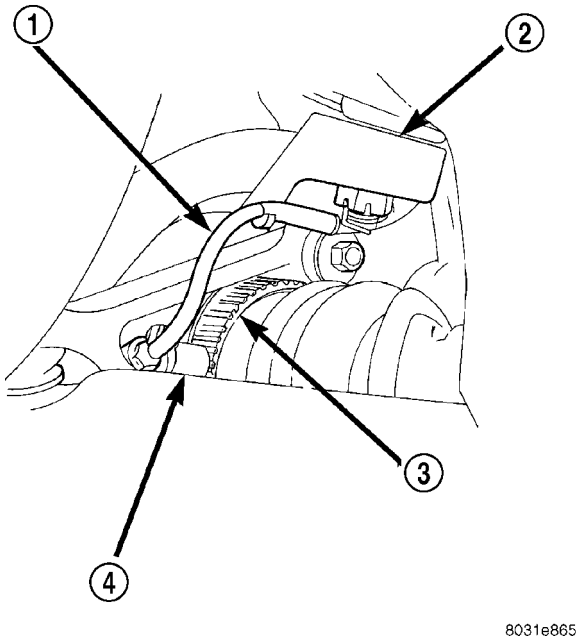

Fig. 2 Front Wheel Speed Sensor

1 - WHEEL SPEED SENSOR PIGTAIL
2 - STEERING KNUCKLE
3 - TONE WHEEL
4 - FRONT WHEEL SPEED SENSOR

INSTALLATION

(1) If **original** sensor will be installed, wipe all traces of old spacer material off sensor pickup face. Use a dry shop towel for this purpose.

(2) Apply Mopar Lock N' Seal or Loctite ® 242 on bolt that secures sensor in steering knuckle. Use new sensor bolt if original bolt is worn or damaged.